

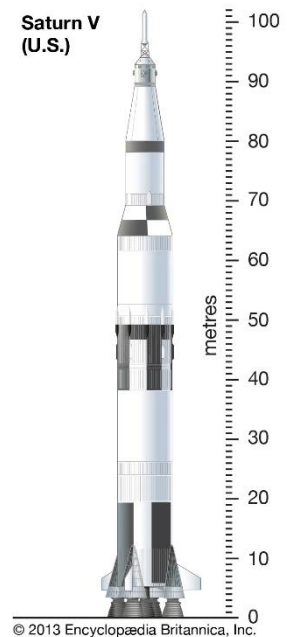
Air Rocket Vertical Launch

The Big Question: How Do We Get to Another Planet?

Synopsis: Remember, all the kinematics we learn help to answer the Big Question above. We must launch a rocket towards a habitable planet (provided we know where it is). However, what keeps us stuck to the ground is the force of gravity, which perpetually pulls us toward the center of the Earth, and if there were no ground, we would continue to accelerate toward it. The rocket propulsion must *completely* overcome Earth's pull to reach another planet – this is not an easy feat, which is why rocket science is challenging, mostly due to the engineering solutions needed to counter the colossal pulling force of Earth. Notice the word "completely". Most amateur vertical launch attempts to reach space fail because the pull of gravity eventually wins over the propulsion, which does not completely overcome it but only delays the inevitable outcome (falling back). Hence the maxim "what goes up, must come down."

Newton was the first to calculate the engineering solution to break this maxim – he showed that a rocket could keep going "up," escaping Earth's pull and not falling back, if launched *fast enough*. So, the key is the launch SPEED. The rocket can escape Earth's gravity with a high enough initial velocity from its propulsion. This velocity is known as the "escape velocity" and is around 11 km/s (11,000 m/s, 24,500 miles per hour). At this speed, you go around the Earth's equator in an hour.

Apollo 11 Moon lander



In this note, we launch an air rocket to model a vertical launch. The basic physics of vertical launch is the same; whether you launch an Apollo mission or a toy rocket, it is all about how fast you thrust the rocket upward. Instead of powerful chemical propellants in real rockets (so an explosion propels the rocket up), an air rocket uses pressurized air released at once, like a darting balloon.

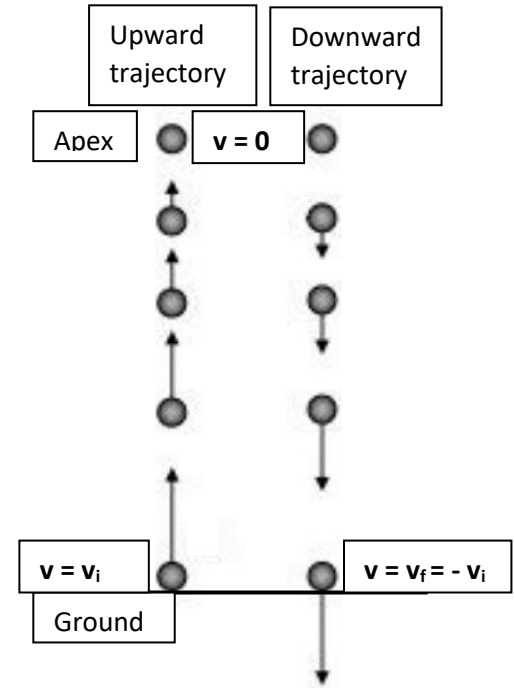
The key equation is $v_f = v_i + at$, where v_f is the final velocity (m/s), v_i is the initial velocity (m/s) which is the focus of this lab, a is the acceleration (m/s^2) which is equal to $a = -g = -9.8 \text{ m/s}^2$, and t is the time of flight (s). In this lab, we determine the air rocket's initial velocity, v_i , by measuring the flight time, *i.e., the time the rocket stays in flight from launch to falling back to Earth*. We then estimate the maximum height

reached by the rocket using the equation, $v_f^2 = v_i^2 + 2ad$, where d is the maximum height (m) and the initial velocity just determined (what should be the final velocity, here?).

Performance Expectations: Launch air rockets with four levels of propulsion (low, medium, high, and super) and measure the corresponding flight time. Determine the initial launch velocities using the velocity equation above and the flight as an input. Determine the maximum height reached by the rocket.

Preamble

Answer the following questions. Check your answers before moving on.



- Gravity is an ever-present force pulling things downward. Recall that a force is an "agent of change" in the form of acceleration. So, this gravity will accelerate things at $a = -g = -9.8 \text{ m/s}^2$. We DEFINE upward direction as positive. Explain, thus, the meaning of the negative sign in $a = -g$ ($g = 9.8 \text{ m/s}^2$ is a magnitude): The negative sign means the upward motion must _____ against the gr_____ force because this force pulls everything do_____, the *opposite* direction to upward motion. When a motion *fights* the force, their *directions* must be opposite.
- Suppose it took 6 seconds from the launch to falling back. How long did it take to reach the apex? _____ s. Explain why/how you got this number.
- When the rocket momentarily reaches the apex, its velocity is zero. Why? _____
- From the apex back to the ground, the motion is as if you played the film of the upward motion in reverse – it's a "mirror image (see figure on the right)" but in the opposite direction. Therefore, when considering motion spanning only the launch to the apex (upward trajectory), explain why the equation for the initial velocity is $v_i = g(0.5t)$, where t is the *flight time* (from the launch to falling back).
- The circle represents the rocket. What do the arrows represent? Answer in three parts: The length represents s____d. The arrowhead indicates d____n. The arrow as a whole represents v____y.
- In the above question, how is s____d different from v____y? V____y has a d____n, but s____d does not.
- The velocity of the rocket when it falls back to the ground is $v = v_{f,\text{ground}} = -v_i$. Explain why i) $v_{f,\text{ground}}$ has the same numerical value:
ii) $v_{f,\text{ground}}$ is the negative of v_i .
- If the initial velocity is 30 m/s , what is $v_{f,\text{ground}}$? _____ m/s .
- The flight time is 6 s . Calculate the initial velocity and the maximum height reached. Show all work.

Instructions

- 1) Following my instructions, launch the air rocket using four levels of thrust strength: low, medium, high, and super. Start with low.
- 2) Measure the flight time: Start the clock as soon as the cap holding the pressure pops off, launching the rocket, and stop the clock when it falls back to the ground. Repeat flight time measurement three times for each level of thrust strength.
- 3) Average the flight times for each level and use this averaged value for all calculations.
- 4) Complete the Flight Time Table below.

Thrust Strength	Flight Time (s)			Average Flight Time (s)
	Trial 1	Trial 2	Trial 3	
Low				
Medium				
High				
Super				

- 5) Using the average flight time, calculate the initial velocity for each level of thrust strength. First, show an example calculation for High Thrust Strength:

Average Flight Time = _____ s.

Equation to use: $v_f = v_i + at$. Solve for v_i . What should v_f be at the apex? What value of t should you use, therefore?

$v_i =$ _____ (Show full expression)

$v_i =$ _____ m/s. **Check your calculations with me before moving on.**

- 6) Reminder: Only the initial velocity determines how high the rocket rises or how long it stays in flight. Now that we have the initial velocity, we calculate the maximum height reached by the rocket.

Equation to use: $v_f^2 = v_i^2 + 2ad$. Solve for d . Set $a = -g = -9.8 \text{ m/s}^2$. What should v_f be at the apex?

First, show an example calculation for High Thrust Strength:

$d =$ _____ (Show full expression)

$d =$ _____ m/s. **Check your calculations with me before moving on.**

7) Complete the maximum height table below.

Thrust Strength	Initial velocity (m/s)	Max. Height (m)
Low		
Medium		
High		
Super		

In Calculus Notation

$a = dv/dt = -g$ (constant). Integrating once gives velocity as a function of time; since $d = \int v \, dt$, integrating that same function again gives height:

$$v(t) = v_i - \int g \, dt = v_i - gt \quad d = \int v(t) \, dt$$

Show both integration steps below: